

Technique 5: Motion Combination

MP

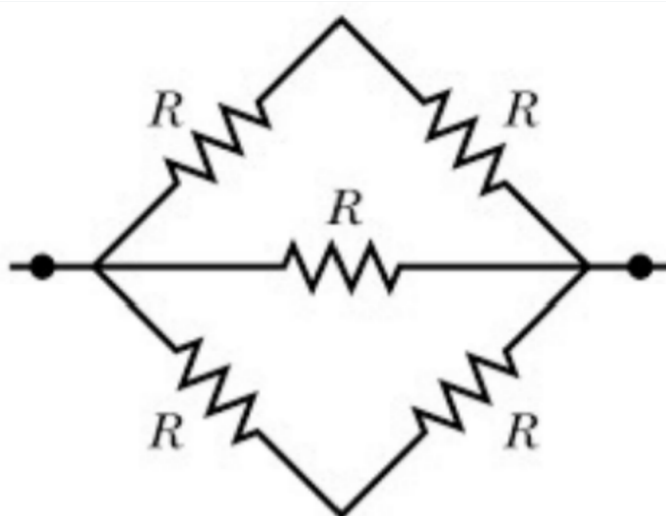
Teach You Physics Hand In Hand

June 2026

1 Solution to Problem in Symmetry Post

Level of difficulty: **3/10**.

a) Consider symmetry, point C and D have same voltage so there is no potential difference across them. The network can be simplified into this:



And the equivalent resistance across A and B is:

$$\frac{1}{\frac{1}{2R} + \frac{1}{R} + \frac{1}{2R}}$$

$$= \frac{R}{2}$$

15 points

b) Since there is no potential difference across C and D, there is no current.

5 points

2 Motion Combination

There are several types of motion combinations:

- Circular motion + constant velocity parallel to circular motion = drifting
- Circular motion + constant velocity perpendicular to circular motion = helix path motion
- Circular motion + instantaneous radial impulse = radial harmonic motion in circular path

The rest of this kind of combined motion requires your modeling skills and they can only be developed by solving a lot of problems.

3 Example

Question:

Consider a charge $+q$ with mass m in a space where there is constant downward electric field E , and the magnetic field B is pointing into paper. The charge is released from rest. If it is required to reach a point that is horizontally L away from the starting point of same vertical height, find the required condition. Assume no gravity.

Answer:

Level of difficulty: **3/10**.

The velocity v is a combination of velocity needed for circular motion v_1 and constant-velocity motion v_2 . The drifting motion is now broken down into two motions superimposed.

$$qE = qv_2B$$

$$v_2 = \frac{E}{B}$$

Time for reaching the point:

$$T = \frac{LB}{E}$$

Period of circular motion:

$$t = \frac{2\pi m}{qB}$$

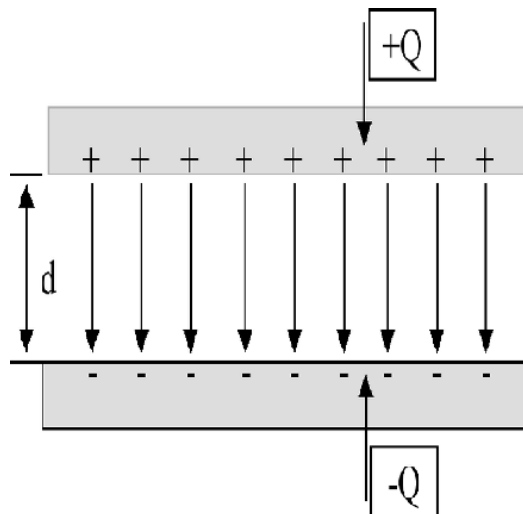
In order to reach a point, the following condition is required:

$$n \frac{2\pi m}{qB} = \frac{LB}{E}$$

where n is a positive integer.

4 Problem

Capacitor and Electrical Breakdown



Consider a circular parallel-plate capacitor of radius R and plate separation d (with $d \ll R$). The capacitor is initially charged to $+Q_0$ on the upper plate and $-Q_0$ on the lower plate. The plates are ideal conductors, and the space between them is vacuum. At $t = 0$, the two plates are connected by a thin straight conducting wire of negligible radius along their common central axis (the z -axis). The wire has resistance r_w and negligible self-inductance. The capacitor discharges through this wire. You may assume the discharge is quasi-static (i.e., the electric field between the plates remains approximately uniform at all times), and neglect edge effects.

Throughout the problem, use cylindrical coordinates (r, ϕ, z) with the z -axis along the wire. Let $Q(t)$ be the charge on the upper plate, and $I_{\text{total}}(t)$ the total conduction current in the wire. The capacitance is $C = \epsilon_0 \pi R^2 / d$, and the time constant is $\tau = r_w C$.

Section A: Introduction

A.1 [2 points] Derive $Q(t)$ and $I_{\text{total}}(t)$ from the RC discharge equation. Express your answer in terms of Q_0, τ, t .

A.2 [4 points] Show that the radial conduction current $I_{\text{cond}}(r, t)$ crossing an imaginary cylindrical surface of radius r toward the axis is:

$$I_{\text{cond}}(r, t) = I_{\text{total}}(t) \frac{r^2}{R^2}$$

A.3 [1 point] Find the magnetic field between the plates and state its direction.

Section B: Fundamental Mechanics

A single electron (charge $-e$, mass m , $e > 0$) is released from rest at point P on the upper plate, at a radial distance r_0 from the central axis ($0 < r_0 < R$).

B.1 [2 points] Write the electric field $\vec{E}(t)$ (magnitude and direction) between the plates. Hence, write the full Lorentz force $\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$ in cylindrical coordinates, resolving it into r, ϕ, z components. The electron has zero initial ϕ direction velocity: $v_\phi(0) = 0$.

B.2 [3 points] From the ϕ -component of the equation of motion, prove that the z -component of angular momentum $L_z = mr^2\dot{\phi}$ is conserved. Hence show that the motion reduces to the 2D coupled system in the (r, z) plane, i.e. find $\ddot{r}$ and $\ddot{z}$.

Section C: Circular Motion and Drift Motion

For the remainder of the problem, we assume the discharge is slow enough that the total current can be taken as approximately constant during the electron's flight time. Furthermore, the radial displacement of the electron is small, so we evaluate the magnetic field at $r = r_0$ and treat it as a constant. We also take the electric field as constant: $E(t) \approx E_0$. The electron is released from rest at $(r_0, 0)$ on the upper plate.

C.1 [4 points] With these approximations, write the coupled equations for $\dot{r} = u$ and $\dot{z} = w$ as a first-order linear system. Solve this system using the initial conditions $u(0) = 0, w(0) = 0$.

C.2 [4 points] Decompose the displacement into a circular motion and a drift motion. Identify the radius of the circle and its center. Verify that it matches the classical $\vec{E} \times \vec{B}/B^2$ drift.

The solution will be uploaded in the next post and MP love you all :)